KEMENTERIAN PENDIDIKAN TINGGI
JABATAN PENDIDIKAN POLITEKNIK DAN KOLEJ KOMUNITI

POLITEKNIK MALAYSIA
BALIK PULAU

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

DFK20013 : WEB DESIGN TECHNOLOGY

LECTURER'S NAME	**PUAN NUR NABILAH BINTI NOORAHIM**	
TYPE OF ASSESSMENT	**PROBLEM BASED ASSIGNMENT**	
TOPIC	3.0 JAVASCRIPT FUNDAMENTALS AND INTERACTIVITY WITH BOOTSTRAP 4.0 INTRODUCTION TO WEB AND MOBILE FRAMEWORK	
DURATION	6 HOURS (NF2F)	
DATE OF ASSESSMENT	WEEK 14 (20/04/2026 - 24/04/2026)	
GROUP	**NAME**	**REGISTRATION NO.**
	IRFAN AIMAN BIN ZURAIDI	21DIT25F1411
	MUHAMMAD HAZEEM SHAHPUTRA BIN MARWAS	21DIT25F1202
	MUHAMMAD ADAM HARIS BIN HASAN	21DIT25F1367
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MARKS	**CLO 3A**	**/50**

Disemak oleh:

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POLITEKNIK MALAYSIA
BALIK PULAU

DFK20013 : WEB DESIGN TECHNOLOGY

PROBLEM BASED ASSIGNMENT

CLO 3A	Demonstrate a good practice and ethics in performing the responsive web and mobile application.	A4, PLO11

OBJECTIVE:

By the end of this assignment, students will be able to:

- Identify weaknesses in a web system and explain their impact on user experience (A2).
- Propose responsible and user-centered improvements for a responsive mobile web application (A3).
- Develop a responsive mobile web prototype using Bootstrap 5 and JavaScript while demonstrating responsible and professional development practice (A3).
- Integrate and justify ethical and professional development decisions in mobile web applications (A4).

TOOLS:

Students are required to use:

1. Sublime Text / Notepad++
2. Web Browser (Google Chrome recommended)
3. Bootstrap 5 via official CDN
4. Microsoft Word / PDF editor (for documentation)

INSTRUCTION:

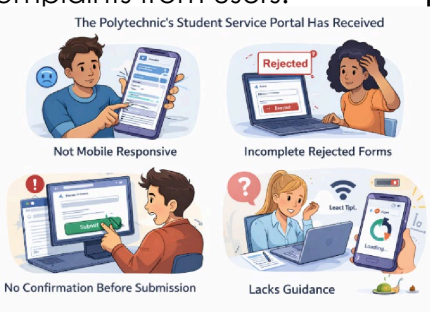
1️⃣ ***Group Work Requirement***

- This Problem-Based Assignment (PBA) must be completed in groups of **3–4 students**.

2️⃣🧩🧩 ***Situation***

The Polytechnic's Student Service Portal has received complaints from users:

- The system is not mobile responsive.
- Students frequently submit incomplete form
- There is no confirmation before submission.
- The interface lacks guidance and clarity.
- The system performs poorly on mobile devices.



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Section A

1. Identify at least three weaknesses of the current portal.

-The system is not mobile responsive.

-no confirmation before submission.

- interface lacks guidance and clarity.

2. Explain how these weaknesses affect users on mobile devices.

it will reduce effectiveness of the website if using it at mobile because of it is not same while using desktop view and mobile view . So, it will affect the mobile user that using website that didnt support mobile view .There are some feature that cant be use with maximum capability.

3. Describe how mobile-first responsive design can improve the system.

First, it improves **user experience**. Since many users access systems using mobile devices, designing for smaller screens ensures the interface is simple, clear, and easy to use. Users don't need to zoom or struggle to navigate, which makes the system more user-friendly.

Second, it increases **performance and speed**. Mobile-first design focuses on loading only essential content first. This reduces unnecessary elements, so the system loads faster, especially on slower internet connections.

Third, it enhances **accessibility**. By prioritizing simple layouts, readable text, and touch-friendly buttons, the system becomes easier for all users to interact with, including those with limited technical skills.

Section B

1.Propose the key features of your redesigned Mobile Web Application

The redesigned mobile web application should include a responsive layout that adjusts to different screen sizes for different user like desktop view,tablet view and mobile view.other than that, a simple navigation menu such as home and help for easy access, fast loading speed through optimized content, secure user authentication, a touch-friendly interface with large buttons, a search and filter function to help users find information quickly, and a notification system to inform users about updates.

2. Explain why Bootstrap 5 is suitable for mobile web development.

Bootstrap 5 is suitable for mobile web development because it follows a mobile-first approach, provides a responsive grid system that adapts to different devices, includes many pre-built components like forms and navigation bars, does not rely on jQuery which improves performance, and ensures compatibility across different web browsers.

3. Explain the importance of client-side validation in improving system reliability.

Client-side validation is important because it ensures that users enter valid data before submitting forms, provides instant feedback to improve user experience, reduces unnecessary load on the server, and increases system reliability by minimizing errors and maintaining accurate data.

4. Explain how interactive components (modal, tooltip, popover, carousel) improve user experience and enhance usability.

Interactive components such as modals, tooltips, popovers, and carousels improve user experience by making the system more engaging and easier to use, as modals allow important information to be displayed without changing pages, tooltips provide helpful hints, popovers give additional details, and carousels present content in an attractive and interactive way.

5. Briefly explain the difference between Mobile Web and Mobile Web Application, and state which category your system belongs to.

Mobile web refers to a standard website accessed through a mobile browser that is usually static, while a mobile web application is more interactive and functions like an app within the browser, and this system belongs to a mobile web application because it includes features like login, validation, and dynamic content.

SECTION C – JavaScript Implementation

Variables and Data Types

```
// ========== FORM VALIDATION (Green ticks / Red errors) ==========
let totalSubmissions = 0;
let currentFeedbackData = { name: "", matrix: "", email: "", comment: "" };
```

At least one condition statement

```
    // Double again for smoother loop
    const newItems = Array.from(track.children);
    if (newItems.length < 15) {
        items.forEach(item => {
            const clone2 = item.cloneNode(true);
            track.appendChild(clone2);
        });
```

At least one loop

```
    const items = Array.from(track.children);
    // Clone items for seamless infinite loop
    items.forEach(item => {
        const clone = item.cloneNode(true);
        track.appendChild(clone);
    });
```

Two functions

```
24  function isMatrixOnlyDigits(value) {
25      return /^\d+$/.test(value);
26  }
27
28  function isValidEmailFormat(email) {
29      return email.includes('@') && email.includes('.') && email.length > 5;
30  }
```

Event handling

```
97   // Add event listeners for real-time validation
98   if (fullNameInput) fullNameInput.addEventListener('input', validateFieldRealTime)
99   if (matrixInput) matrixInput.addEventListener('input', validateFieldRealTime);
100  if (emailInput) emailInput.addEventListener('input', validateFieldRealTime);
101
```

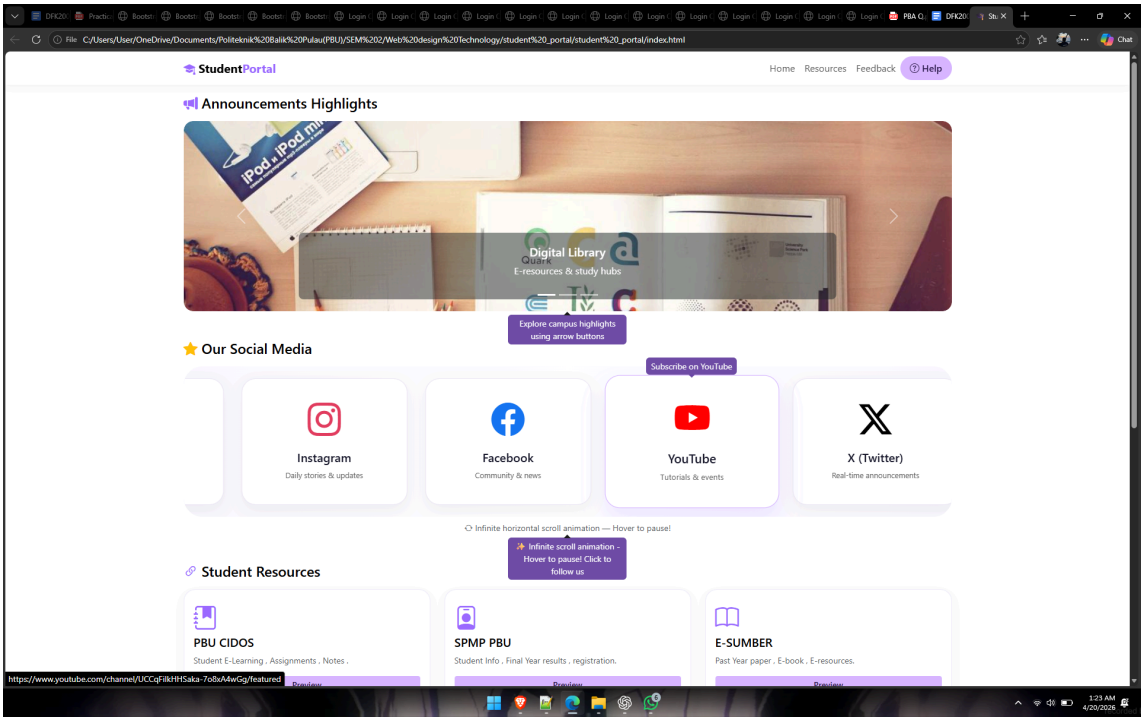
Section E

1. Explain how responsiveness, validation, and interactivity work together as an integrated mobile web solution.

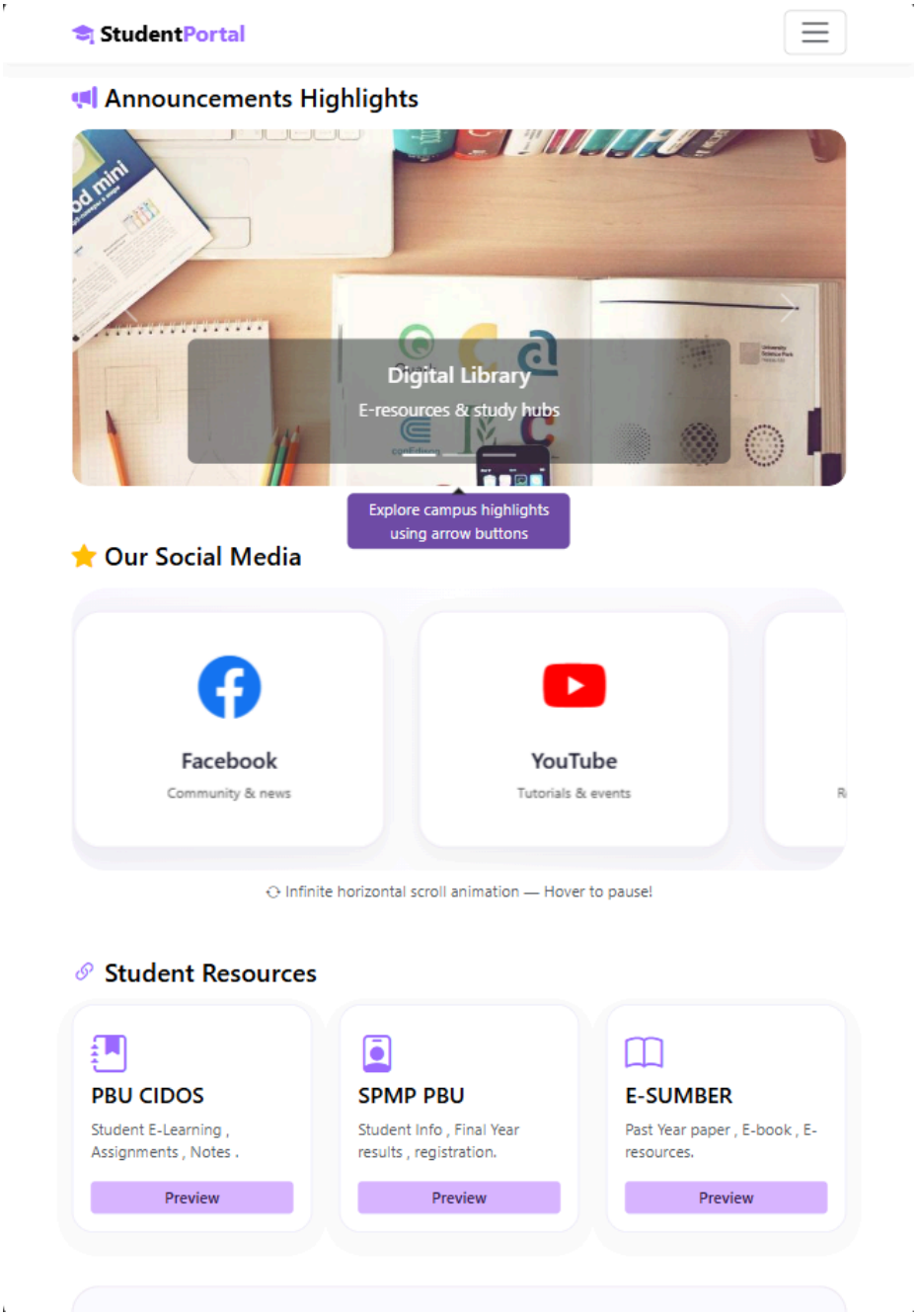
Responsiveness, validation, and interactivity work together to make a mobile web system better and easier to use. Responsiveness makes sure the website fits all screen sizes, validation checks user input to avoid errors, and interactivity helps users understand and use the system easily. Together, they make the system smooth, accurate, and user-friendly.

2. Test your application in:

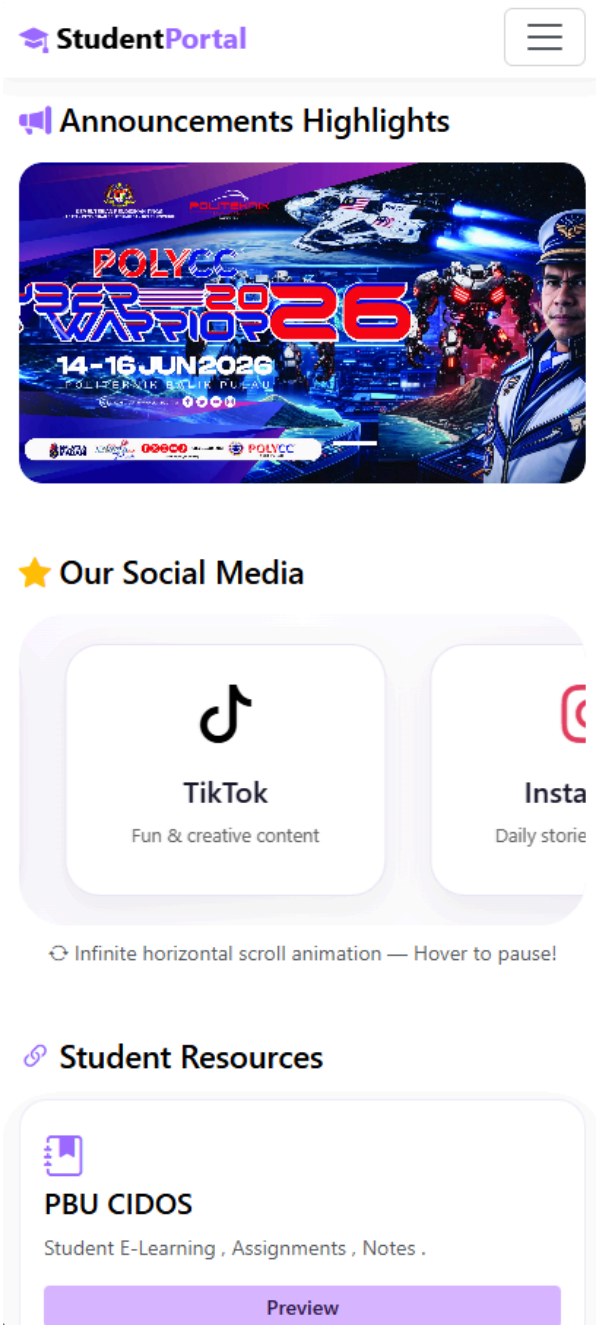
o Desktop view



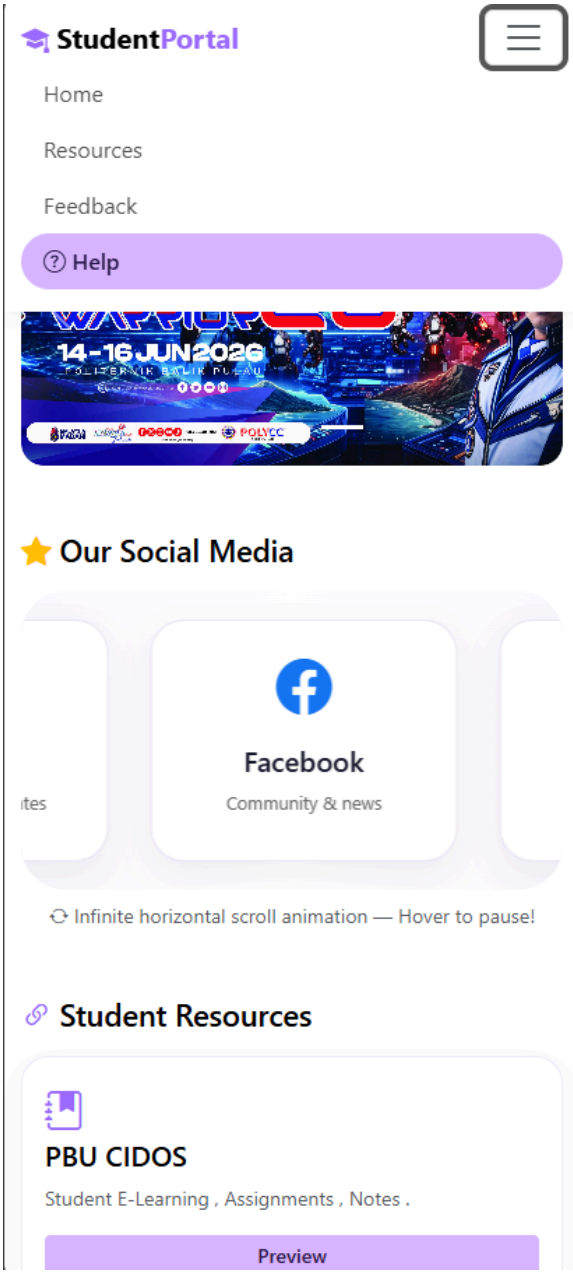
o Tablet view

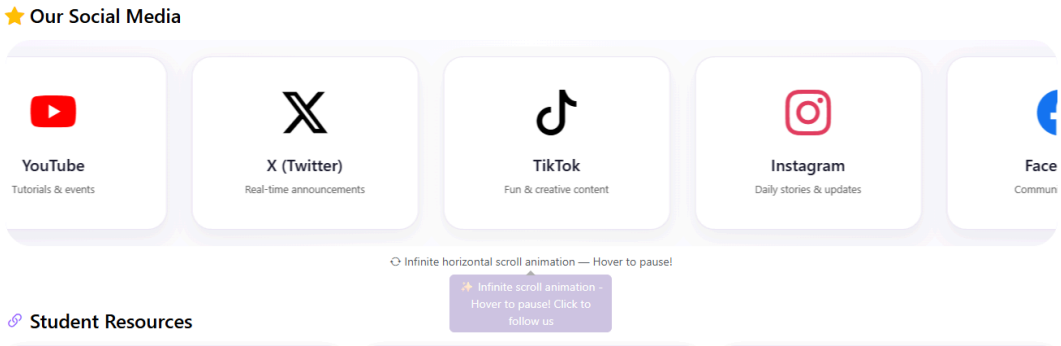


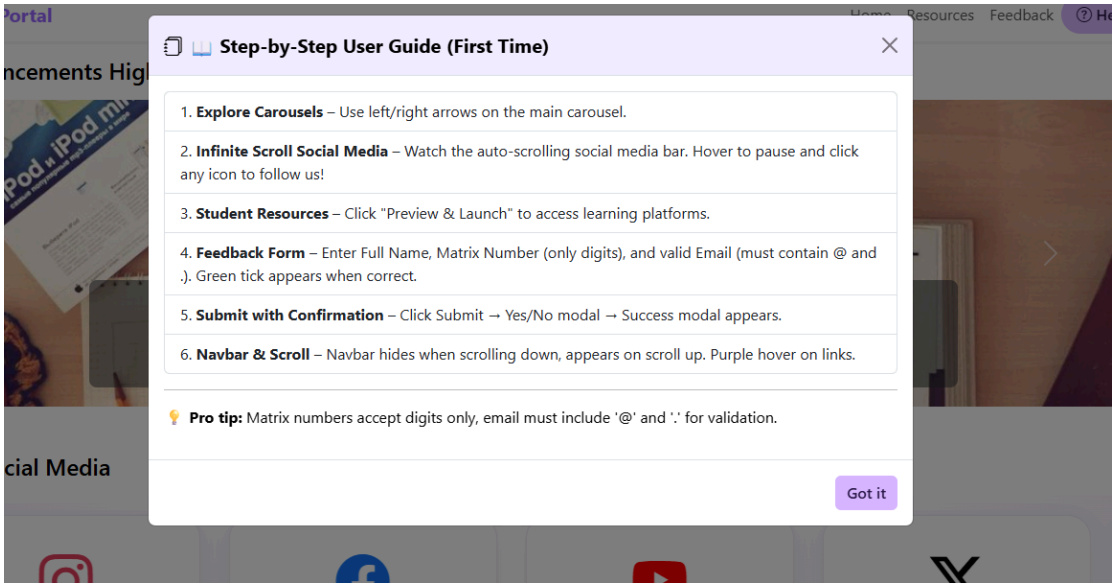
o Mobile view



3. Provide screenshots. Reflect on how your redesigned portal improves the original portal.






Portal
Resources
Feedback
ncements Hig
cial Media
Step-by-Step User Guide (First Time)
1. Explore Carousels – Use left/right arrows on the main carousel.
2. Infinite Scroll Social Media – Watch the auto-scrolling social media bar. Hover to pause and click any icon to follow us!
3. Student Resources – Click "Preview & Launch" to access learning platforms.
4. Feedback Form – Enter Full Name, Matrix Number (only digits), and valid Email (must contain @ and .). Green tick appears when correct.
5. Submit with Confirmation – Click Submit → Yes/No modal → Success modal appears.
6. Navbar & Scroll – Navbar hides when scrolling down, appears on scroll up. Purple hover on links.
Pro tip: Matrix numbers accept digits only, email must include '@' and '.' for validation.
Got it

Explanation of development decisions (Section C & D report)

SECTION C

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/js/bootstrap.bundle.min.js"></script>
<!-- External JavaScript -->
<script src="js/script.js"></script>
```

SECTION D

1. Bootstrap 5 integrated via official CDN.

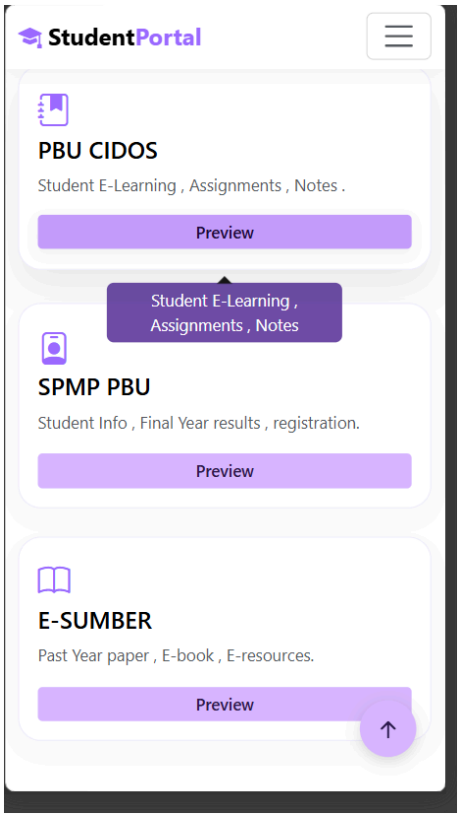
```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0, viewport-fit=cover">
<title>Student Portal | Infinite Social Scroll</title>
<!-- Bootstrap 5 CDN + Icons -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/css/bootstrap.min.css" rel="stylesheet">
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/bootstrap-icons.min.css">
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0, viewport-fit=cover">
  <title>Student Portal | Infinite Social Scroll</title>
  <!-- Bootstrap 5 CDN + Icons -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/css/bootstrap.min.css" rel="stylesheet">
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/bootstrap-icons.min.css">
  <!-- External CSS -->
  <link rel="stylesheet" href="css/style.css">
<!-- Favicon - Browser Tab Logo -->
<link rel="icon" type="image/x-icon" href="data:image/svg+xml,<svg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 100 100'><text y='.9em' font-size='90'>🎓</text></svg>">
</head>
<body>

<!-- ========= NAVBAR ========= -->
<nav class="navbar navbar-expand-lg fixed-top" id="mainNavbar">
  <div class="container">
    <a class="navbar-brand fw-bold" href="#"><i class="bi bi-mortarboard-fill" style="color:#9b6bff;"></i> Student<span style="color:#9b6bff;">Portal</span></a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarMain">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarMain">
      <ul class="navbar-nav ms-auto mb-2 mb-lg-0">
        <li class="nav-item"><a class="nav-link" href="#home">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#student-links">Resources</a></li>
        <li class="nav-item"><a class="nav-link" href="#feedback">Feedback</a></li>
        <li class="nav-item"><a class="nav-link btn-purple-light rounded-pill px-3" href="#" data-bs-toggle="modal" data-bs-target="#userGuideModal"><i class="bi bi-question-circle"></i> Help</a></li>
      </ul>
    </div>
  </div>
</nav>

<!-- Scroll to top button -->
<div class="scroll-top-btn" id="scrollTopBtn" aria-label="Scroll to top">
  <i class="bi bi-arrow-up-short"></i>
</div>

<main class="container my-5 pt-4" id="home">

  <!-- SECTION 1: Main Carousel -->
  <section class="mb-5" id="sec-carousel" data-bs-toggle="tooltip" data-bs-placement="bottom" title="Explore campus highlights using arrow buttons">
    <h2 class="h4 fw-semibold mb-3"><i class="bi bi-megaphone-fill" style="color:#9b6bff;"></i> Announcements Highlights</h2>
    <div id="mainCarousel" class="carousel slide" data-bs-ride="carousel">
      <div class="carousel-indicators">
        <button type="button" data-bs-target="#mainCarousel" data-bs-slide-to="0" class="active"></button>
        <button type="button" data-bs-target="#mainCarousel" data-bs-slide-to="1"></button>
        <button type="button" data-bs-target="#mainCarousel" data-bs-slide-to="2"></button>
      </div>
      <div class="carousel-inner rounded-4 overflow-hidden">
        <div class="carousel-item active"><img src="https://picsum.photos/id/20/1200/400" class="d-block w-100" alt="Library"><div class="carousel-caption d-none d-md-block bg-dark bg-opacity-50 rounded-3"><h5>Digital Library</h5><p>E-resources & study hubs</p></div></div>
        <div class="carousel-item"><img src="img/bg_cidos_lms.png" class="d-block w-100" alt="Campus"><div class="carousel-caption d-none d-md-block bg-dark bg-opacity-50 rounded-3"><h5>Student & Lecterur Lounge</h5><p>Center for eLearning and Teaching, CeLT</p></div></div>
        <div class="carousel-item"><img src="img/poster.jpg" class="d-block w-100" alt="Graduation"><div class="carousel-caption d-none d-md-block bg-dark bg-opacity-50 rounded-3"><h5>Upcoming Events</h5><p>Academic</p></div></div>
      </div>
      <button class="carousel-control-prev" type="button" data-bs-target="#mainCarousel" data-bs-slide="prev"><span class="carousel-control-prev-icon"></span></button>
      <button class="carousel-control-next" type="button" data-bs-target="#mainCarousel" data-bs-slide="next"><span class="carousel-control-next-icon"></span></button>
    </div>
  </section>
```

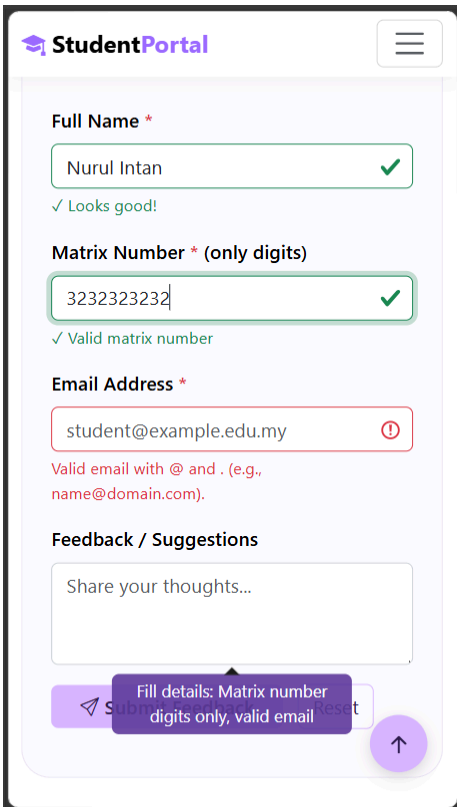
```html
<!-- SECTION 2: INFINITE SCROLL ANIMATION - Social Media -->
<section class="mb-5" data-bs-toggle="tooltip" data-bs-placement="bottom" title="✨ Infinite scroll animation - Hover to pause! Click to follow us">
  <h2 class="h4 fw-semibold mb-3"><i class="bi bi-star-fill text-warning"></i> Our Social Media</h2>
  <div class="social-infinite-wrapper">
    <div class="social-track" id="infiniteSocialTrack">
      <div class="social-item" data-bs-toggle="tooltip" title="Follow us on Instagram @studentportal">
        <a href="https://www.instagram.com/politeknikbalikpulau/" target="_blank">
          <i class="bi bi-instagram" style="color:#e4405f;"></i>
          <h5>Instagram</h5>
          <p>Daily stories & updates</p>
        </a>
      </div>
      <div class="social-item" data-bs-toggle="tooltip" title="Like our Facebook page">
        <a href="https://www.facebook.com/polibalikpulau" target="_blank">
          <i class="bi bi-facebook" style="color:#1877f2;"></i>
          <h5>Facebook</h5>
          <p>Community & news</p>
        </a>
      </div>
      <div class="social-item" data-bs-toggle="tooltip" title="Subscribe on YouTube">
        <a href="https://www.youtube.com/channel/UCCgFilkHHSaka-7o8xA4wGg/featured" target="_blank">
          <i class="bi bi-youtube" style="color:#ff0000;"></i>
          <h5>YouTube</h5>
          <p>Tutorials & events</p>
        </a>
      </div>
      <div class="social-item" data-bs-toggle="tooltip" title="Follow us on X (Twitter)">
        <a href="https://x.com/Poli_BalikPulau" target="_blank">
          <i class="bi bi-twitter-x" style="color:#000000;"></i>
          <h5>X (Twitter)</h5>
          <p>Real-time announcements</p>
        </a>
      </div>
      <div class="social-item" data-bs-toggle="tooltip" title="Follow our TikTok">
        <a href="https://www.tiktok.com/@politeknikbalikpulau" target="_blank">
          <i class="bi bi-tiktok" style="color:#010101;"></i>
          <h5>TikTok</h5>
          <p>Fun & creative content</p>
        </a>
      </div>
    </div>
  </div>
  <div class="text-center mt-2 small text-muted">
    <i class="bi bi-arrow-repeat"></i> Infinite horizontal scroll animation — Hover to pause!
  </div>
</section>

<!-- SECTION 3: Student resources -->
<section id="student-links" class="mb-5">
  <h2 class="h4 fw-semibold mb-3" data-bs-toggle="tooltip" title="Click Preview & Launch to access resources"><i class="bi bi-link-45deg" style="color:#9b6bff;"></i> Student Resources</h2>
  <div class="row g-4">
    <div class="col-md-4" data-bs-toggle="tooltip" title="Student E-Learning , Assignments , Notes">
      <div class="card h-100 p-3">
        <i class="bi bi-journal-bookmark-fill fs-1" style="color:#9b6bff;"></i>
        <h5>PBU CIDOS</h5>
        <p class="text-muted small">Student E-Learning , Assignments , Notes .</p>
        <a href="https://pbulms.cidos.edu.my/" target="_blank" class="btn btn-purple-light btn-sm">Preview </a>
      </div>
    </div>
    <div class="col-md-4" data-bs-toggle="tooltip" title="Student Info , Final Year results , registration">
      <div class="card h-100 p-3">
        <i class="bi bi-person-badge fs-1" style="color:#9b6bff;"></i>
        <h5>SPMP PBU</h5>
        <p class="text-muted small">Student Info , Final Year results , registration.</p>
        <a href="http://spmp.pbu.edu.my/" target="_blank" class="btn btn-purple-light btn-sm">Preview </a>
      </div>
    </div>
    <div class="col-md-4" data-bs-toggle="tooltip" title="Past Year paper , E-book , E-resources">
      <div class="card h-100 p-3">
        <i class="bi bi-book fs-1" style="color:#9b6bff;"></i>
        <h5>E-SUMBER</h5>
        <p class="text-muted small">Past Year paper , E-book , E-resources.</p>
        <a href="https://sites.google.com/view/e-sumberpbu/utama" target="_blank" class="btn btn-purple-light btn-sm">Preview</a>
      </div>
    </div>
```

```html
    </div>
  </section>

  <!-- SECTION 4: Feedback Form -->
  <section id="feedback" class="mb-5" data-bs-toggle="tooltip" title="Fill details: Matrix number digits only, valid email">
    <div class="feedback-section p-4 p-md-5">
      <h3 class="h4 fw-semibold mb-4"><i class="bi bi-chat-dots-fill" style="color:#9b6bff;"></i> Student Feedback Form</h3>
      <form id="feedbackForm" novalidate>
        <div class="row g-3">
          <div class="col-md-4">
            <label class="form-label fw-medium">Full Name <span class="text-danger">*</span></label>
            <input type="text" class="form-control" id="fullName" placeholder="Nur Aina Binti Ahmad" required>
            <div class="error-feedback" id="nameError">Name required.</div>
            <div class="valid-feedback-custom" id="nameValid">✓ Looks good!</div>
          </div>
          <div class="col-md-4">
            <label class="form-label fw-medium">Matrix Number <span class="text-danger">*</span> (only digits)</label>
            <input type="text" class="form-control" id="matrixNo" placeholder="20231234567" required>
            <div class="error-feedback" id="matrixError">Matrix number must contain only digits (0-9).</div>
            <div class="valid-feedback-custom" id="matrixValid">✓ Valid matrix number</div>
          </div>
          <div class="col-md-4">
            <label class="form-label fw-medium">Email Address <span class="text-danger">*</span></label>
            <input type="email" class="form-control" id="email" placeholder="student@example.edu.my" required>
            <div class="error-feedback" id="emailError">Valid email with @ and . (e.g., name@domain.com).</div>
            <div class="valid-feedback-custom" id="emailValid">✓ Email format correct</div>
          </div>
          <div class="col-12">
            <label class="form-label fw-medium">Feedback / Suggestions</label>
            <textarea class="form-control" id="feedbackMsg" rows="3" placeholder="Share your thoughts..."></textarea>
          </div>
          <div class="col-12">
            <button type="button" class="btn btn-purple-light px-4" id="openConfirmBtn"><i class="bi bi-send"></i> Submit Feedback</button>
            <button type="reset" class="btn btn-outline-purple ms-2" id="resetFormBtn">Reset</button>
          </div>
        </div>
      </form>
      <div id="formStatusMsg" class="mt-3 small"></div>
    </div>
  </section>
</main>

<!-- FOOTER -->
<footer class="pt-4 pb-3 mt-3">
  <div class="container">
    <div class="row gy-3">
      <div class="col-md-5">
        <h6 class="fw-bold"><i class="bi bi-question-circle-fill"></i> Help & Support</h6>
        <ul class="list-unstyled small">
          <li><a href="#" data-bs-toggle="modal" data-bs-target="#userGuideModal">📘 Step-by-step User Guide</a></li>
          <li><a href="tel:+60123456789" data-bs-toggle="tooltip" title="Click to call support"><i class="bi bi-telephone-fill"></i> +60 12-345 6789 (click to call)</a></li>
          <li><a href="mailto:support@studentportal.edu.my" data-bs-toggle="tooltip" title="Send email"><i class="bi bi-envelope-fill"></i> support@studentportal.edu.my</a></li>
        </ul>
      </div>
      <div class="col-md-4 text-center">
        <i class="bi bi-mortarboard fs-2" style="color:#9b6bff;"></i>
        <p class="mb-0 mt-2 small">Student Portal<br>Empowering digital campus</p>
      </div>
      <div class="col-md-3 text-md-end">
        <p class="small mb-0">© 2026 Student Portal | All rights reserved</p>
        <p class="small text-muted">Mobile-first | Bootstrap 5</p>
      </div>
    </div>
  </div>
</footer>
```

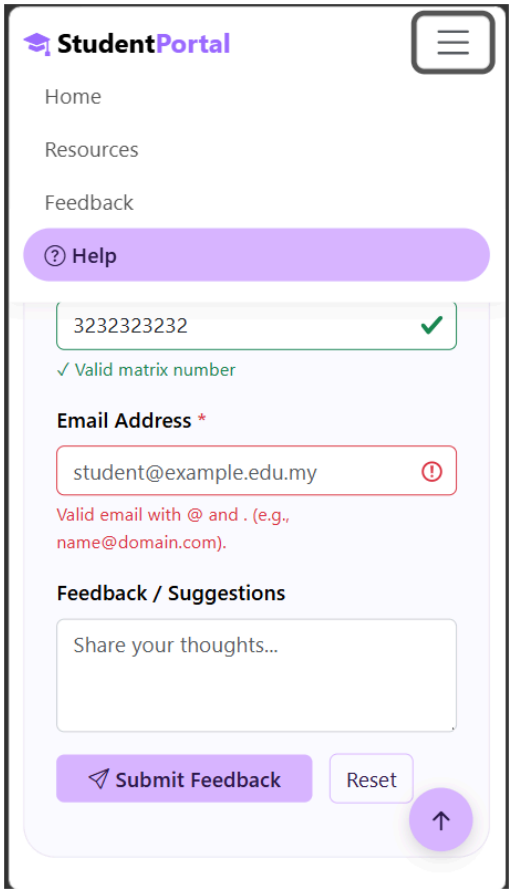
Responsive layout using Bootstrap Grid (3 columns stacking vertically on smaller screens).



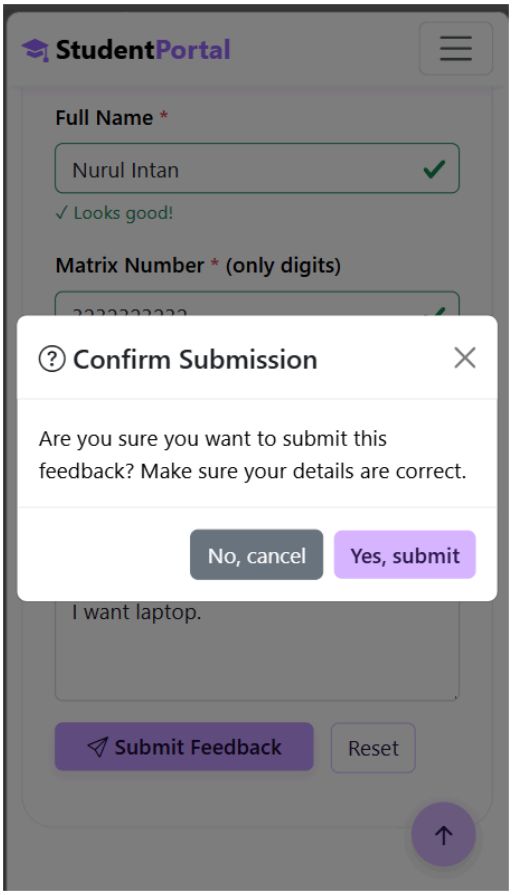
Client-side validation



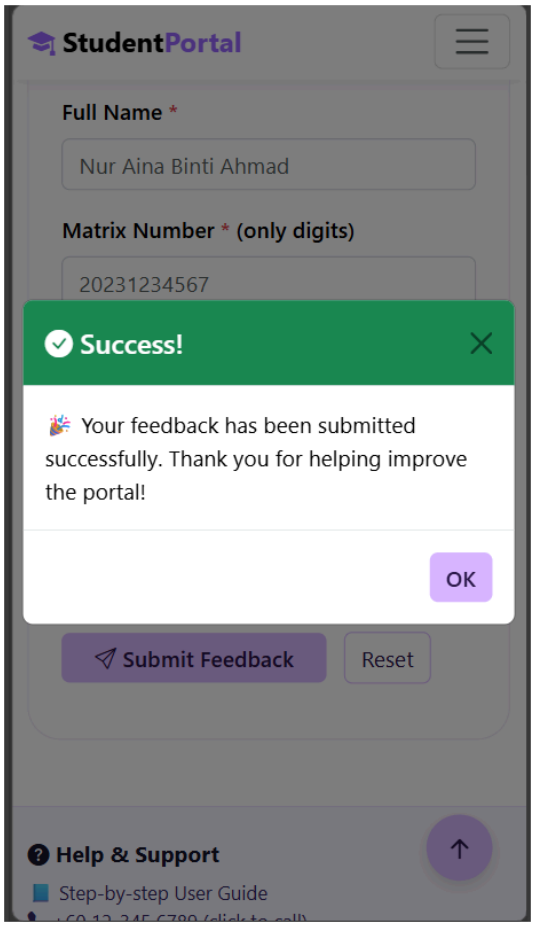
Navbar



Modal (confirmation dialog)



Popover



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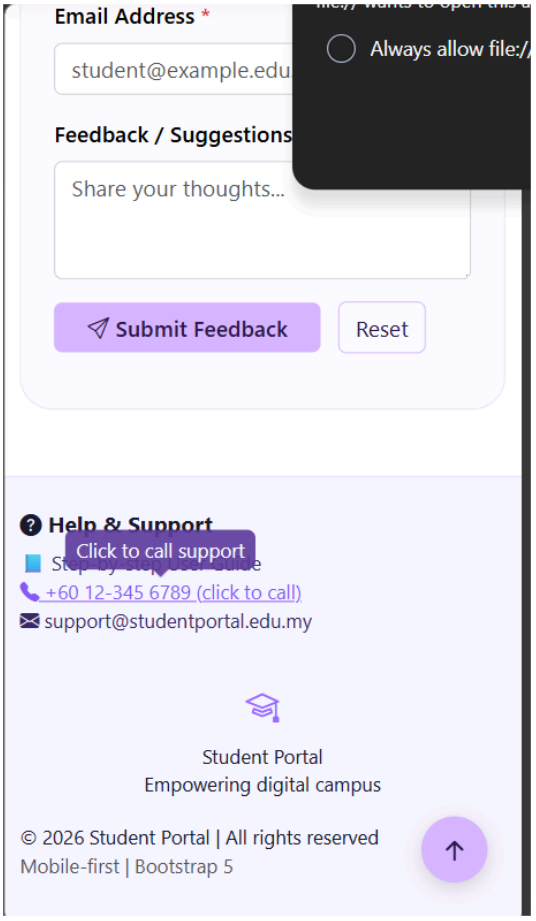

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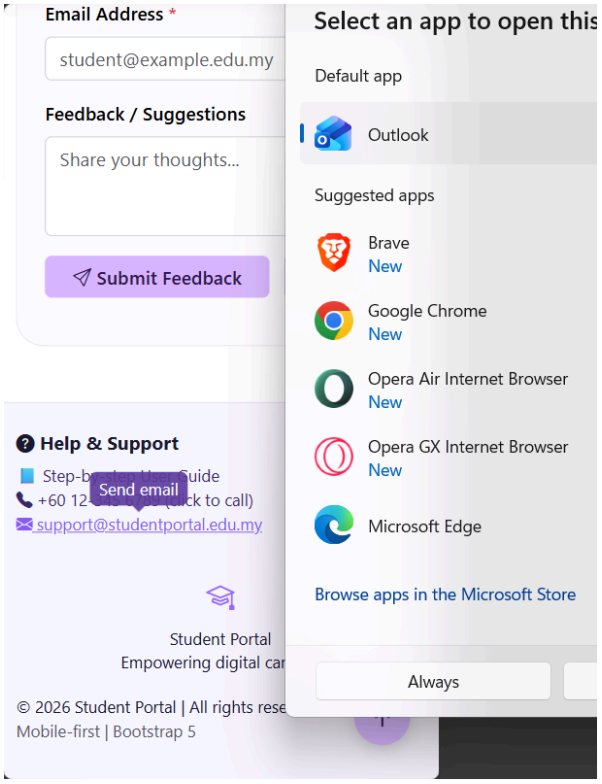

CAROUSEL

Explanation of Development Decisions

1. Responsible Development Practice

Responsible development ensures the application is reliable, secure, and considerate of the user's time and data.

Decision Made	How it Reflects Responsible Practice
Real-time Client-Side Validation (validateFieldRealTime with green ticks/red errors)	Error Prevention: By providing instant feedback on input format (digits only for Matrix No, @ and . for Email), we prevent the user from submitting invalid data. This reduces server load and user frustration compared to a page reload with generic errors.
Explicit Confirmation Modal (checkFormAndStore opens confirmSubmitModal)	Ethical Interaction: The user is not tricked into a final submission. The modal forces a deliberate second click ("Confirm Submit") which aligns with ethical design principles—giving users control and preventing accidental actions.

2. Structured Development Practice

Structured development ensures the code is maintainable, scalable, and bug-resistant.

Decision Made	How it Reflects Structured Practice
Separation of Concerns (CSS in /css, JS in /js)	Maintainability: By keeping style, logic, and markup separate, another developer (or your future self) can easily locate and update code without breaking the layout.
Functional Decomposition (Breaking code into small functions)	Reusability & Readability: Instead of one giant submit function, the logic is split into: isMatrixOnlyDigits, validateFieldRealTime, checkFormAndStore, and finalSubmitFeedback. This modular structure makes the code easier to test and debug.
Looping for Setup (forEach for cloning carousel items)	Efficiency: Instead of hardcoding 10+ slides for the infinite scroll, the decision to use cloneNode in a loop demonstrates an understanding of DRY (Don't Repeat Yourself) programming. It dynamically generates the required elements, making the HTML cleaner and the application easier to scale.

3. User-Centered Development Practice

User-centered design prioritizes the goals, tasks, and limitations of the end-user (in this case, a student on a mobile device).

Decision Made	How it Reflects User-Centered Practice
Implementation of "Scroll to Top" Button (scrollTopBtn)	Navigation Efficiency: On a long portal page, scrolling back to the navbar manually is tedious on mobile. The floating button is a standard mobile UX pattern that allows users to quickly reset their position, improving navigation flow.
Seamless Infinite Scroll Animation (Carousel clone loop)	Engagement & Aesthetics: A static list of announcements is easily ignored. A smoothly looping animation of updates/achievements captures attention without requiring user clicks. This is a micro-interaction designed to make the portal feel dynamic and "alive."
Bootstrap Tooltip Integration (data-bs-toggle="tooltip")	Contextual Help: For complex terms or buttons (e.g., "Matrix No Format"), a tooltip provides just-in-time help. This prevents the user from having to leave the form to search for instructions, keeping them focused on the task.
Real-Time Validation with Visual Icons (Green checkmark / Red cross)	Cognitive Ease: Users don't have to guess what the format should be. The green checkmark provides positive reinforcement immediately upon correct entry. This reduces the cognitive load compared to an error message appearing only after pressing Submit.

ASSESSMENT'S RUBRIC

COURSE CODE: DFK20013

COURSE NAME: WEB DESIGN TECHNOLOGY

TYPE OF ASSESSMENT: PROBLEM BASED ASSIGNMENT

NO	REGISTRATION NO.:	NAME:
1	21DIT25F1411	IRFAN AIMAN BIN ZURAIDI
2	21DIT25F1202	MUHAMMAD HAZEEM SHAHPUTRA BIN MARWAS
3	21DIT25F1367	MUHAMMAD ADAM HARIS BIN HASAN
4	21DIT25F1090	MUHAMMAD HANIF FAUZAN BIN YUSMAN

Criteria	5	4	3	2	1	Weighted	Marks
📘📘 **SECTION A – Problem Analysis (A2) – 5 Marks**							
Identification, Impact & Improvement	Clearly identifies ≥3 relevant weaknesses, explains mobile impact logically, and clearly links responsive design & validation as solutions.	Identifies ≥3 weaknesses with good explanation but limited linkage depth.	Identifies 2 weaknesses with basic explanation.	Identifies 1 weakness or explanation unclear.	Weak or generic statements.	×1	/5
📘📘 **SECTION B – Value-Based Proposal (A3) – 15 Marks**							
Feature Proposal, Bootstrap Justification & Concept Differentiation	Comprehensive proposal with strong professional reasoning; clearly justifies Bootstrap 5, validation importance, interactive components, and	Good proposal and justification with minor conceptual gaps.	Basic proposal with limited reasoning.	Weak justification; partial understanding.	Very minimal reasoning.	×3	/15

	accurately differentiates Mobile Web vs Mobile Web Application.						
SECTION C – JavaScript Implementation (A3) – 10 Marks							
Variables, Condition, Loop, Functions & Event Handling	All required JS elements correctly implemented, functional, logically structured and integrated.	Minor functional or syntax errors.	Most elements present but incomplete.	Limited JS logic; missing key elements.	Minimal or incorrect logic.	×2	/10
SECTION D – Mobile Web Development & Integration (A3) – 15 Marks							
Responsive Layout, Bootstrap Components, Validation & Phone Integration	Fully responsive layout; Bootstrap CDN correct; Modal, Tooltip, Popover, Carousel functional; validation working; tel & mailto functional.	Minor responsiveness or interaction issues.	Most components working but incomplete responsiveness.	Several components not functional; weak mobile adaptation.	Minimal implementation.	×3	/15
SECTION E – Professional Integration & Testing (A4) – 5 Marks							
Integration Explanation, Testing & Reflection	Clear integration explanation; complete Desktop, Tablet & Mobile screenshots; strong reflection.	Good explanation with complete screenshots.	Basic explanation; incomplete testing evidence.	Weak integration explanation.	Very minimal reflection.	×1	/5
SECTION F – Documentation, Creativity & Professional Practice (A3) (5 Marks)							
Report Structure, Code Neatness, File Organization & UI	Well-structured report; clean and properly indented code; organized folders; professional and creative UI design.	Minor formatting or structure issues.	Acceptable documentation; readable but inconsistent.	Poor organization; messy code.	Very minimal documentation.	×1	/5
						Total Marks	**/50**